

BSCS FINAL PROJECT

Software Requirements Specification

PLAYERSZWZ

AI-Powered Sports Matchmaking & Arena Booking
Application



Product Owner

Mr. Asif Farooq

Presented by:

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Faculty of Information Technology & Computer Science

University of Central Punjab

Software Requirements Specification

Version: 1.2

Product Owner:

Mr. Asif Farooq

Scrum Master

Afaq Ul Islam

S26CS024

Team Roles

Member Name	Roles
Waleed Abid	Mobile app architecture, backend integration and project coordination
Afaq Ul Islam	Scrum Master, Frontend implementation, maps/location services and UI flows
Hassan Ahmed	AI recommendation module, testing, documentation and QA

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Revision History & Sprint Log

Version	Sprint	Date	Changes	Owner
1.0	Initial Planning	10 April 2026	Project idea and Phase 1 SRS draft prepared from approved proposal.	Waleed Abid
1.1	Template Correction	27 April 2026	Formatting, alignment, diagrams, and irrelevant sections revised according to supervisor feedback.	Afaq Ul Islam
1.2	Agile SRS Update	07 May 2026	Agile framework, backlog, user stories, sprint plan, Scrum activities, acceptance criteria, and IV & V details aligned with the new template.	Hassan Ahmed

Previous Phases Feedback

Idea Defense Feedback (Screenshot)



Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	CS-78
Project Title	PLAYERZWZ

Overall ok.

Your idea has been approved by the committee. Your next milestone is **Phase-I (RS)**. You are advised to start working on it.

The idea defense feedback confirms that the project title PLAYERSZWZ was approved and the team was advised to continue work on Phase-I (RS).

Abstract

PLAYERSZWZ is an intelligent app that aims to reduce the hassle involved in organizing casual sports matches on a daily basis. Everyone wants to play cricket, football, futsal, badminton or alike games, be it students, office professionals or locality players. Often, informal matches are delayed or cancelled as team members are not complete, players confirm very late and arena/ground booking is done with haphazard calls or WhatsApp group. Designed mobile application system offered to facilitate the discovery of players, the creation/ of games join requests, arena booking, notifications, feedback and reliability tracking under a single mobile workflow. Users will sign up, keep sports preferences, find nearby players, create games, join games, ask for arena slot and get reminders / arena booking updates. An AI-powered recommendation module will sequentially rank the relevant games and players for each user by combining factors such as sport preference, proximity, availability, past participation and reliability score. This project demonstrates geolocation services, recommendation logic, mobile computing, cloud database design, and backend API writing. The project will minimize the time spent on coordination and enhance trust between unfamiliar players, which will result in a much higher frequency of participation in local sports. When implemented, the Project will provide a powerful software solution in an academic context.

1. Introduction and Background

1.1 Product Vision & Problem Context

The goal of PLAYERSZWZ is to simplify, quicken and make trustworthy the process of planning for sports for those who often face inconvenience of getting enough players/ booking a slot to play. At present, the procedure is that the game organizer will send a message to several groups, wait for responses, call friends up, and still be unsure if every player who agrees will make it in time. This is also true for arenas, where the information is so scattered that a player might not be sure which ground he may be able to get at the preferred time. Through a structured platform on mobile, PLAYERSZWZ can resolve this problem wherein players can discover players on their flight (nearby), publish games with open spots, join relevant games, and request bookings for an arena. In the same way, it is possible to implement a system of reliability levels which will help make a better decision before accepting any unknown player in a match. The admin, university students, young working professionals, casual players, game organizers, and arena managers are included users. We are not looking to create yet another social sports app. Our main objective is to reduce the coordination friction and make the whole journey of finding players and booking a ground more reliable.

User Research Approach

To validate the needs from which the design can fulfil, the team intends to do a small survey and short interviews with some users. You can divert to other area wherein you can realize a fictive image of wherever on look. Questions supplement interview will ask how do they currently set up games.

- What results in cancellations?
- What are some ways to trust unknown players?
- What information do you need in order to join a game?

Expected advantages include the ability for participants to discover suitable games in a shorter time-span which will result in lesser cancellations, for organizers creation of a team made easier + becoming more visible for join requests, for arena owners and managers who will be able to get a clearer picture of booking demand which will result in lesser manual chasing, academic work is expected to involve: Flutter and Flask APIs, Firebase, Maps, notifications, and recommendation logic.

1.2 Stakeholders Identification

Stakeholder	Role	Expectations
Player	A main mobile application user searching for games or possible teammates in the nearby vicinity.	Simple sign-up, relevant suggestions, clear game description, safe participation, reliable notifications.
Game Organizer	Establishes Match, Sets Sport, Time and Location. Accept or deny entry requests. Could begin reservation procedure.	This team is quick to fill, reliable participants, easy approval, see participant count.
Arena Manager / Owner	Arenas details for maintenance. Reply to the bookings request.	Less phone coordination and easier slot management due accurate booking request (1st and 2nd).
Administrator	Assess reports and control abuse. Act on flagged accounts. Check the quality of the platform etc.	History of reports cleared, access based on role, audit trail, option for warning/suspension.
Project Supervisor / Product Owner	Having authority over the project scope. Confirms specifications. Evaluates work.	Following the SRS template, the scope of work is feasible, the requirements are measurable, and the phase 1 was a success.
Development Team	Create mobile application. Create backend APIs. Construct recommendation module.	Prioritized backlog, defined sprint plan and roles.

2. Agile Framework Setup

Scrum will be used for the product as there are a number of modules which will be developed, reviewed and improved in short iterations. The Project Owner will be the Project Manager who will represent the stakeholder perspective in the project development team. Scrum Master will be the group lead. The responsibility of the development team includes mobile development, backend integration, the AI logic, documentation and testing. The sprint time will be calculated to enable the increment to be meaningful and enough time to revise and refine 2 weeks.

Scrum Role	Assigned Person / Group	Main Responsibility
Product Owner	Mr. Asif Farooq	Authorizes the scope, orders the feedback, evaluates the major deliverables.
Scrum Master	Waleed Abid	Organizes sprints, tracks tasks, keeps the team on schedule.
Development Team	Waleed, Afaq, Hassan	Develops, tests, documents and improve.

The tools that you'll be using are task tracking tools (Trello or Jira), version control (Git/GitHub), and UI wireframes (Figma). Short sprint discussions are conducted on WhatsApp/Google Meet and firebase console is used to set up authentication/databases.

2.1 Epics For the Project

Epic ID	Epic Name	Description	Related Problem Area
EP-01	User Account Management	Handles registration, login, logout, verification, session handling, and role-based access.	Users need secure access before using matchmaking and booking features.
EP-02	Player Profile & Preferences	Allows players to maintain sport interests, area, availability, radius, bio, and reliability score.	The system needs accurate user data to make discovery and recommendations useful.
EP-03	Nearby Discovery	Finds players and games near a selected location using map/location filters.	Manual searching wastes time and causes delays when arranging matches.
EP-04	Game Management	Supports game creation, join requests, participant approval, game status, and participant tracking.	Organizers need a structured way to complete teams and manage confirmations.
EP-05	Arena Discovery & Booking	Displays nearby arenas, available slots, booking requests, and booking status updates.	Arena booking is often manual and not visible to players in one place.
EP-06	AI Recommendation Engine	Ranks players and games using sport preference, proximity, availability, history, and trust score.	Users need relevant suggestions instead of browsing every option manually.
EP-07	Feedback, Rating & Moderation	Captures ratings, reports, no-show cases, reliability impact, and admin decisions.	The platform needs trust controls to reduce misuse and unreliable participation.
EP-08	Notification System	Sends alerts for join requests, booking updates, reminders, cancellations, and approvals.	Users miss updates when coordination is scattered across different channels.

2.2 Product Backlog

Backlog ID	Epic	Feature Description	Priority
PB-01	User Account Management	Implement user registration using email, password, city, and preferred sport.	High
PB-02	User Account Management	Develop secure login, logout, account verification, and session handling.	High
PB-03	Player Profile & Preferences	Create profile management for sports, bio, availability mode, and search radius.	High
PB-04	Nearby Discovery	Implement nearby player discovery using location permission and geohash/radius filters.	High
PB-05	Game Management	Allow organizers to create games with sport, date, time, venue, and required players.	High
PB-06	Game Management	Allow users to view open games and request to join.	High
PB-07	Game Management	Allow organizers to approve or reject join requests.	High
PB-08	Arena Discovery & Booking	Display arena list with sport type, location, and available slots.	High
PB-09	Arena Discovery & Booking	Submit booking request and track pending/approved/rejected status.	High
PB-10	AI Recommendation Engine	Provide rule-based recommendations for new users using sport and area.	Medium
PB-11	AI Recommendation Engine	Improve recommendations using activity history and reliability score.	Medium
PB-12	Feedback, Rating & Moderation	Allow players to rate or report participants after a completed game.	Medium
PB-13	Feedback, Rating & Moderation	Calculate and display reliability score using no-shows, ratings, and reports.	Medium
PB-14	Notification System	Send push notifications for join requests, approvals, booking status, and reminders.	Medium
PB-15	Admin Moderation	Create admin panel for reviewing reports and account actions.	High
PB-16	Quality & Testing	Prepare test cases, IV & V log, and regression testing checklist.	Medium

2.3 User Stories

Story ID	Backlog ID	User Story	Priority	Acceptance Criteria	Focal Person
US-01	PB-01	As a new user, I want to register an account so that I can access sports matchmaking features.	High	Valid data creates account; duplicate email shows error.	Waleed
US-02	PB-02	As a registered user, I want to log in and log out securely so that my account remains protected.	High	Valid credentials open dashboard; invalid attempts show clear error.	Waleed
US-03	PB-03	As a player, I want to update sports preferences and availability so that others can find me correctly.	High	Saved profile updates reflect in discovery and recommendations.	Afaq
US-04	PB-04	As a player, I want to discover nearby players so that I can complete a team faster.	High	List/map shows compatible players within selected radius.	Afaq
US-05	PB-05	As an organizer, I want to create a game so that interested players can join it.	High	Game is saved, visible nearby, and has correct player count.	Waleed
US-06	PB-06	As a player, I want to join an open game so that I can participate without manual calls.	High	Join request is recorded or immediate join is confirmed.	Afaq
US-07	PB-07	As an organizer, I want to approve or reject join requests so that I can control team quality.	High	Participant list updates and user receives notification.	Waleed
US-08	PB-08	As a user, I want to browse arenas near me so that I can choose a suitable place to play.	High	Arena list shows location, sports, and available slots.	Afaq
US-09	PB-09	As an organizer, I want to send a booking request so that a slot can be reserved formally.	High	Booking is stored with Pending status and manager/admin is notified.	Waleed
US-10	PB-10	As a new user, I want basic suggestions so that I can still find games without history.	Medium	Rule-based results are shown using sport and location.	Hassan
US-11	PB-11	As a regular user, I want personalized suggestions so that I see the most relevant games first.	Medium	Recommendations are ranked using preference, distance, activity, and reliability.	Hassan
US-12	PB-12	As a player, I want to rate or report someone after a game so that the platform remains trustworthy.	Medium	Feedback is stored once per participant per game.	Hassan
US-13	PB-14	As a user, I want notifications so that I do not miss requests, approvals, or game reminders.	Medium	Push alert and in-app log are generated for defined events.	Afaq
US-14	PB-15	As an admin, I want to review reports so that misuse can be handled fairly.	High	Admin can warn, suspend, or clear reported accounts with logs.	Hassan

2.4 Story Points & Estimation

Story ID	User Story (Short)	Complexity Level	Story Points	Justification
US-01	User Registration	Medium	3	Requires validation, Firebase Auth, profile storage, and verification handling.
US-02	Login / Logout	Medium	3	Authentication, session handling, failed-attempt management, and logout flow.
US-03	Manage Profile	Medium	3	Profile UI, Firestore update, validation, and recommendation inputs.
US-04	Nearby Players	High	5	Geolocation, filters, radius query, reliability display, and permission fallback.
US-05	Create Game	Medium	3	Form validation, game model, organizer mapping, and visibility logic.
US-06	Join Game	Medium	3	Join request flow, status handling, and game capacity checks.
US-07	Approve Requests	Medium	3	Organizer action, participant update, notification, and rejection handling.
US-08	Arena Discovery	High	5	Map/list integration, slot display, filters, and incomplete data handling.
US-09	Booking Request	High	5	Booking lifecycle, status changes, notifications, and conflict checks.
US-10	Rule-based Recommendation	Medium	3	Fallback algorithm with sport, area, and availability signals.
US-11	AI Recommendation	High	8	Ranking logic, data preparation, history signals, and trust score integration.
US-12	Rating / Report	Medium	3	Feedback form, duplicate prevention, reliability impact, and moderation link.
US-13	Notifications	Medium	5	Firebase Cloud Messaging, in-app log, retries, and reminder scheduling.
US-14	Admin Moderation	High	5	Admin access, report review, action logging, and account status control.

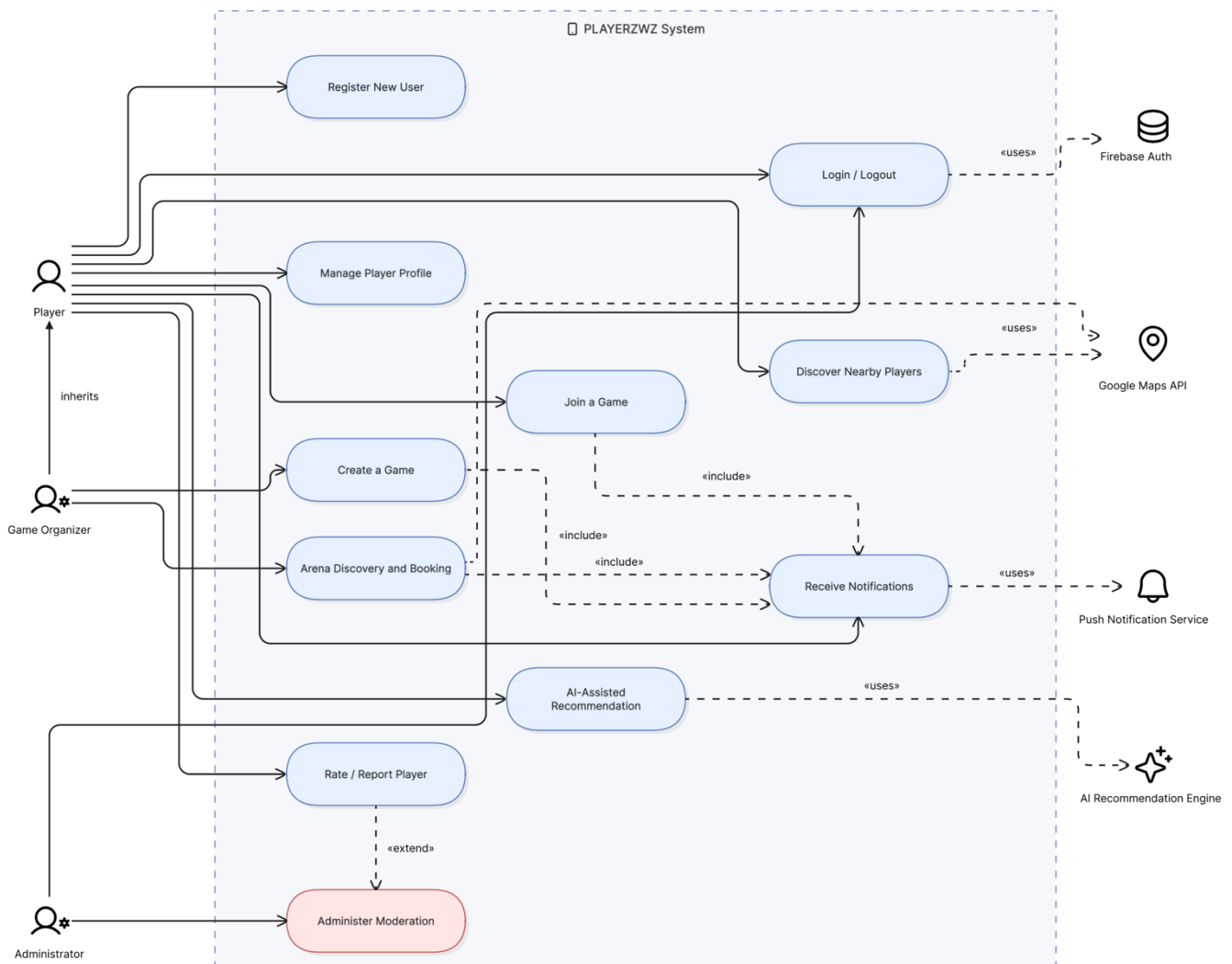
2.5 Sprint Planning & Sprint Backlog

Sprint No.	Duration	Sprint Goal	Selected User Stories	Total Story Points
Sprint 1	2 Weeks	Set up project structure, authentication, base database models, and profile screens.	US-01, US-02, US-03	9
Sprint 2	2 Weeks	Build player discovery, game creation, join requests, and organizer approval flow.	US-04, US-05, US-06, US-07	14
Sprint 3	2 Weeks	Implement arena listing, booking request workflow, notifications, and status updates.	US-08, US-09, US-13	15
Sprint 4	2 Weeks	Implement recommendation logic, rating/report flow, admin moderation, testing, and documentation.	US-10, US-11, US-12, US-14	19

3. Functional Requirements

3.1 Use Case Diagram

The diagram below shows the main actors and system boundaries. The Player utilizes the System Registration as well as Login/Logout, Profile Management, Player Discovery, Join Game, Ratings and Recommendations. The Game Organizer allows players to make bookings or create their game. Administrator checks Reports and Moderation action. The system connects with outside services including Firebase Authentication, Google Maps API, Push Notification Service and the AI.



PLAYERSZWZ

Use Case ID	Use Case Name	Primary Actor	Description	Mapped Story ID	User Story
UC-01	User Registration	New User	Allows a new user to create an account using email, password, city, and preferred sport.	US-01	As a new user, I want to register an account so that I can access the system.
UC-02	User Login / Logout	Registered User	Enables secure sign-in, active session creation, and logout.	US-02	As a registered user, I want to log in securely so that I can access my dashboard.
UC-03	Manage Player Profile	Player	Allows editing of sports preferences, location radius, availability, and bio.	US-03	As a player, I want to update my profile so that other users can find me correctly.
UC-04	Discover Nearby Players	Player	Displays compatible nearby players using sport, area, radius, availability, and reliability filters.	US-04	As a player, I want to discover nearby players so that I can complete a team faster.
UC-05	Create a Game	Game Organizer	Creates a new sports match with sport, time, venue, and required player count.	US-05	As an organizer, I want to create a game so that interested players can join it.
UC-06	Join a Game	Player	Records join requests or immediate participation depending on organizer settings.	US-06	As a player, I want to join an open game so that I can participate without manual calls.
UC-07	Arena Discovery and Booking	Game Organizer / Player	Browses arenas, checks slots, and submits booking requests.	US-08, US-09	As a user, I want to browse arenas and request booking so that a place can be reserved.
UC-08	AI-Assisted Recommendation	Player	Ranks games and players using preference, distance, activity history, and reliability score.	US-10, US-11	As a user, I want relevant suggestions so that I can find suitable games quickly.
UC-09	Rate / Report Player	Player	Stores feedback after a completed game and updates reliability metrics.	US-12	As a player, I want to rate or report someone so that the platform remains trustworthy.
UC-10	Notifications and Updates	System / User	Sends alerts for requests, approvals, reminders, booking status, and cancellations.	US-13	As a user, I want notifications so that I do not miss important updates.
UC-11	Administer Moderation	Administrator	Reviews reports, warns or suspends accounts, and logs decisions.	US-14	As an admin, I want to review reports so that misuse can be handled fairly.

4. Non-functional Requirements

4.1 Reliability

- The system will store user profile, game, booking, rating and notification records in cloud storage with validating prior to storing.
- Normal network interruptions shall be dealt with by the application, displaying readable error messages and attempting time-sensitive operations again, if feasible.
- The state of the join-request and booking shall remain consistent so that no contradictory information about the participant/slot is shown to the users.
- No-show and rating updates will be auditable for the purpose of reliability score changes, if challenged.

4.2 Performance

- According to academic scale, the dashboard should load in less than three seconds on normal home or university WIFI network.
- For a specified radius, the Nearby Player and Open Game search should return the first results within 5 seconds or less.
- During prototype testing, output should occur in 3 seconds for Rule Based Fallback, and 6 seconds for History Based Ranking.
- At least 50 academic test users should be able to use the application concurrently without any noticeable slowdown of the backend.

4.3 Use-ability

- The user must be able to access and register the Dashboard with no technical support.
- It should be easy to either make a game or join a game, and there should be only a few fields.
- The whole app should not be denied access because of the location, but it should allow the user to manually enter the area if possible.
- Labels should be readable; buttons consistent and sport/distance/time filters should be simple.

4.4 Other Non-functional Requirements

Attribute	Requirement Description
Security	Passwords must never be stored in an easily readable form (plain text). Adding Firebase Auth and secure tokens will enable this application to authenticate users and manage session validity.
Maintainability	It is intended that the Mobile UI, backend APIs, recommendation logic, and database access will all be modular in design, so obtaining or featuring new functionality will be easy in future updates.
Scalability	The proposed system architecture should be flexible enough to accommodate future growth in users and additional arenas, additional sports categories, and any other future recommendation signals
Portability	The primary platform for the application will initially be Android; however, the architecture should also support Flutter/iOS through the same codebase at a later date.
Interoperability	The system will allow integration with Firebase, Google Maps/location services, push notifications, and API Integration with the backend.
Ethics & Fairness	Ratings and reliability ratings must be composed of a fair evaluation of confirmed data, allowing for disputes to be reviewed before an unfounded penalty is issued against ratings or reliability ratings based on user feedback.

5. Scrum Activities Plan

The work will be kept visible and manageable using scrum activities. The team will pick up items from the backlog at the beginning of each sprint, agree on who owns the items and establish a small sprint goal. A short daily check-in will be done during the sprint to discuss what has been completed, what is being worked on and what is blocking. Each team will check the increment at the end of each sprint, gather feedback and record the next sprint's improvements.

5.1 Kanban Board / Task Tracking

The team will use a Kanban board to monitor the work, which includes four columns: To Do, In Progress, Testing, and Done. Each task will have a responsible and it will be only moved when it reaches the exit condition.

To Do	In Progress	Testing	Done
Design profile preference screen	Implement login API integration	Test password and email validation	Project structure created
Prepare arena slot model	Build create-game form	Test join request edge cases	Firebase project configured
Write recommendation rule set	Integrate Google Maps filter	Test booking status updates	Basic database schema drafted
Create admin moderation UI	Connect notification triggers	Run regression checklist	SRS template aligned

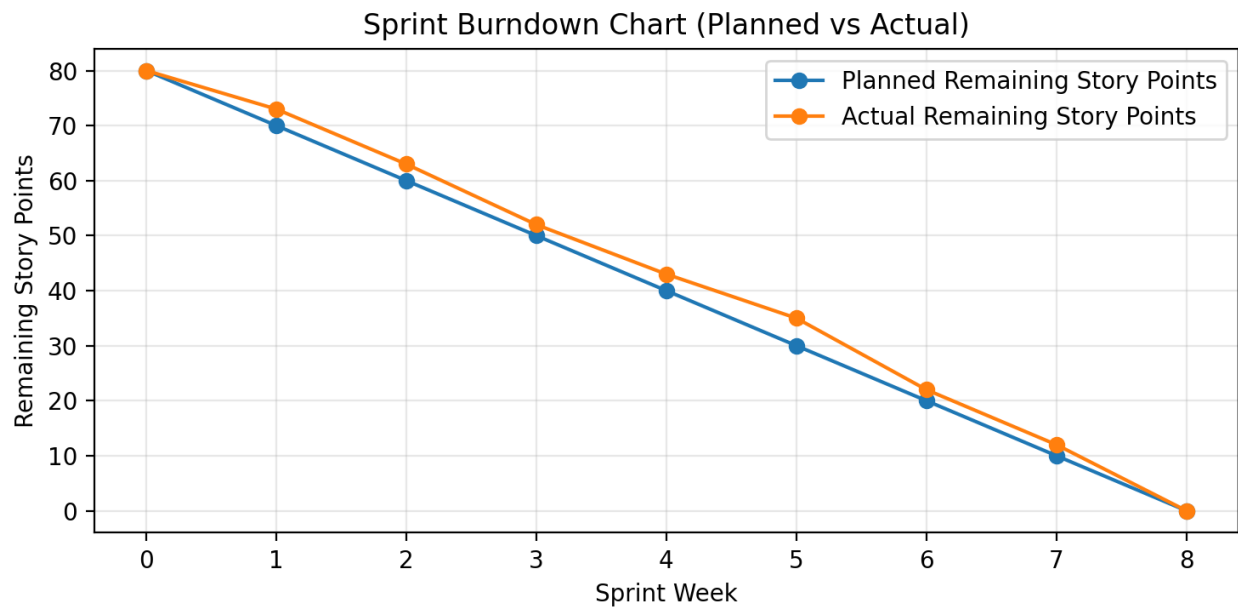
5.2 Acceptance Criteria

Feature	Test Scenario	Expected Result
Registration & Login	User enters valid credentials and receives access to the dashboard.	Valid account opens dashboard; invalid credentials show error without crash.
Profile Management	User updates sport preference, area, availability, and radius.	Updated profile is saved and used in discovery filters.
Player Discovery	User searches nearby players using sport and radius.	Compatible players appear with distance and reliability indicator.
Game Creation	Organizer creates a game with required fields.	Game is visible to eligible nearby users and has correct status.
Join Request	Player taps Join on a suitable game.	Request is saved; organizer receives notification; capacity rules are enforced.
Arena Booking	User selects an arena slot and sends booking request.	Booking status starts as Pending and updates to Approved/Rejected when actioned.
Recommendation	User opens recommendation page/dashboard.	Relevant games/players are ranked and fallback works for new users.
Rating / Report	User submits feedback after completed game.	Feedback is stored once and impacts reliability only through defined rules.
Admin Moderation	Admin reviews report and selects action.	Action is logged and user status updates correctly.

5.3 Release Criteria

Release Area	Criteria
Functional Completion	Any all high-priority user stories (Functional Stories) from Sprints 1-4 have been completed (implemented/demonstratable).
Testing	The following core test cases have passed: Registration, Login, Profile, Discovery, Game, Booking, Recommend, and Moderate.
Data Integrity	There are no open defects regarding duplicate join request, duplicate rating or invalid booking status.
UI Review	Any main flow is intuitive to new users with consistent naming conventions of buttons/labels.
Supervisor Approval	The product owner/supervisor has accepted the phase deliverables following a thorough review.

5.4 Burndown Chart & Progress Tracking



Week	Planned Remaining SP	Actual Remaining SP	Progress Note
0	80	80	Sprint backlog finalized.
2	60	63	Authentication and profile work mostly complete; discovery started.
4	40	43	Game flow completed; arena integration required extra testing.
6	20	22	Recommendations and notifications under testing.
8	0	0	Sprint scope closed and documentation updated.

5.5 Backlog Refinement Plan

Backlog refinement will be held once a week or any time when the priority of the work is changed by the supervisor. The team will discuss open stories, break large stories, fill in missing stories acceptance criteria, estimate new tasks, and remove tasks that are outside the scope of Phase 1. The decisions made about refinement will be written in the sprint log to keep the log consistent with the actual project progress.

6. Risk Analysis

Risk ID	Risk Description	Impact	Mitigation Strategy
R-01	Location permission denied by users	High	Provide manual area entry and explain why location improves nearby results.
R-02	Low activity for new users reduces recommendation quality	Medium	Use rule-based fallback using sport preference, area, and availability until enough history exists.
R-03	Arena slot data becomes inaccurate	High	Mark booking as request-based and require confirmation before showing Approved status.
R-04	Unfair ratings damage reliability score	High	Prevent duplicate ratings, allow dispute window, and reduce score only for confirmed no-show or repeated reports.
R-05	Backend or Firebase integration delays	Medium	Implement core flows first and keep mock data fallback for UI testing.
R-06	Scope becomes too large for Phase 1	High	Keep tournament, commercial payment settlement, and international deployment outside current scope.
R-07	Map API or notification service limitations	Medium	Use documented API limits, fallback messages, and retry logic for notifications.
R-08	Team workload imbalance	Medium	Assign clear owners and review progress during weekly Scrum check-ins.

7. References

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Appendix A: Glossary

Term	Definition
AI Recommendation Engine	A module that matches games and players based on their sport preference, distance from the sport, availability of the player, the player's event history, and the player's reliability score.
Arena	A registered sports facility such as a cricket ground, futsal court or badminton hall.
Availability Mode	A profile flag that indicates if a player is currently available to join games.
Booking	A formal request to hold an arena for the selected date and time.
Cloud Firestore	A Firebase NoSQL database stores all users, profiles, games, bookings, ratings and notifications.
Coordination Friction	Delays in arranging sports games through manual processes and excessive follow-ups.
Game Organizer	A user who creates a game and manages the participant's requests and can create an arena reservation.
Geohash	A technique that indexes geographic locations to group based on coordinates of proximity for faster radius searches.
Join Request	A request sent by a player to join an already existing open game.
No-show	A person confirmed to participate in a game who does not show but has not made a proper cancellation.
Player	The primary user of the application who searches for games, joins games, manages their profile and receives suggestions on other games.
Reliability Score	A numerical indication of the player's dependability and trustworthiness assessed based on attendance, rating, reporting and no-show history.
Scrum	An agile development framework that organizes its development into sprints, testing and continues to evolve.
Sprint	An iterative fixed development period or cycle of two weeks with specific backlog items from which a target/goal can be assessed.
Use Case	A formal description of interaction between an actor and the system to achieve a specific goal.

Appendix B: IV & V Report

(Independent Verification & Validation)

IV & V Resource

Name: Afaq Ul Islam

Signature: _____

S#	Defect Description	Origin Stage	Status	Fix Time Hours	Minutes
1	UC-07 did not include behavior when arena response is delayed or a selected slot becomes unavailable.	Requirements Elicitation	Resolved	0	40
2	Reliability Score was referenced but needed a clearer description for calculation, fairness, and dispute handling.	SRS Drafting	Resolved	1	45
3	Performance requirements used vague wording and were revised with measurable academic-scale thresholds.	SRS Review	Resolved	0	30
4	New template required Agile sections including epics, backlog, user stories, sprint planning, and Scrum activity plan.	Template Alignment	Resolved	1	10

Total Defects Found	Resolved	Pending	Total Fix Time
4	4	0	5 hours 05 minutes

IV & V Notes

All listed defects were reviewed during internal documentation checking before finalizing this SRS. The updated version aligns the project with the new template, improves testability, and adds Agile planning artifacts needed for Phase 1. Appendix B will be expanded in later phases to record implementation defects, test case failures, validation results, and fixes against the requirements defined in this document.